

Associate Director & hands-on technical leader specializing in **high-throughput microservices**, **distributed backend systems**, and **production AI infrastructure**. Architect & manage scalable, sub-second latency, **event-driven microservices** (RabbitMQ, load balancers), & AI-powered developer workflows. Proven expertise in AWS infrastructure (ECS, Postgres, CDK), cost optimization (\$200K+ annual savings), and integrating emerging AI frameworks into **enterprise-scale** production environments.

TECHNICAL EXPERIENCE

Associate Director

May, 2025 — Present

Qure AI

Bangalore, India

- Extracted qCT processing into an event-driven microservice using **RabbitMQ**, meeting strict TAT/SLOs while saving **\$200K+** annually
- Implemented (**MCP**) servers for autonomous **agentic workflows**: built organization-level skill repository with MCP tooling (pr-review.md, deployment.md, test-cases.md, jira-ticket.md) integrated with JIRA APIs for contextual validation and automated test case generation
- Architected **CI/CD AI review** with 3-tier workflow: **JIRA MCP** contextual checks, directory-scoped AI rules (**Claude.md**), & **pytest**
- Designed AI Navigator tagging system (JIRA-style) achieving **sub-500ms** queries across **100+ tags** per patient with efficient bulk operations
- Built AI-powered CI/CD review system with JIRA-context checks, directory-scoped rules, and pytest gating, impacting **50+ developers**
- Own and operate CI/CD infrastructure handling **500+ daily builds** on autoscaling EC2, offloading heavy workloads to GitHub Actions
- Re-architected test infrastructure (EFS, compute, Bazel), reducing test time from 80 min to **20 min** and deployments from 3 hrs to **30 min**
- Migrated from RDS to Aurora Serverless with RDS Proxy, reducing downtime by **90%** and improving high-concurrency resilience

Sr. AI Engineer

June, 2022 — April, 2025

Qure AI

Bangalore, India

- Architected database optimization strategy reducing CPU utilization from **50% to under 20%**, delivering \$16K annual infrastructure cost savings through **query optimization**, **indexing** strategies, and connection management, **explain plan** analysis.
- Designed and deployed secure, user-friendly translation application with **SSO authentication (Keycloak/OAuth2)**
- Migrated UI regression testing from LambdaTest to self-hosted Jenkins fleet on AWS, reducing execution time by **20%** and operational cost by **50%** through infrastructure optimization
- Maintained production **Metabase fork** supporting dashboards in **8+ languages** with enhanced internationalization (i18n) framework

Data Scientist

June, 2021 — May, 2022

Hilabs

Pune, India

- Led **20+ client POCs**, translating business needs into data solutions, contributing to **5 deal closures** and a **multi-year contract**
- Re-engineered report pipeline using **PySpark and Hadoop**, reducing processing time for **1M+ records** from **2 days to 3 hours**
- Built **U.S. physician directory** by aggregating data from **20+ insurance sites** with scalable scraping, error handling, and rate limiting
- Implemented **data quality layer** removing **89K+ invalid addresses**, improving report accuracy and **CRM reliability**

PERSONAL PROJECTS

AI Powered Credit Card Recommendation (cards.aayushostwal.com)

- Modern credit card comparison platform for Indian consumers featuring **50+ cards** with **35+ filterable** parameters.
- Built with **React/TypeScript**, implementing side-by-side comparison & AI-powered personalized recommendation engine using **Groq LLM API**.

Technical Portfolio (aayushostwal.com)

- TypeScript-based personal site deployed on Vercel with content feeds and topic-based filtering for technical writing discoverability.

Medium Author (medium.com)

- Author of 40+ in-depth technical articles published in Data Science Collective, Level Up Coding, and related industry publications, reaching thousands of readers globally.
Topics include: **LLM architecture and AI agents**, **Database optimization**, **Backend systems**

SKILLS

Languages:	Python, TypeScript, SQL, Bash, Groovy, C++, Clojure, Scala, PySpark
Frameworks:	Django, FastAPI, Claude, LangChain, TorchServe, Flask, N8N, PageIndex, Celery, React, Bazel
CI/CD + DevOps:	Jenkins, GitHub Actions, Docker, Kubernetes basics, Git, JIRA integration/automation
Databases & Data	PostgreSQL, MySQL, MongoDB, KnowledgeGraph, Redis, RabbitMQ, Metabase, Grafana, Datadog

EDUCATION

B.Tech., Mechanical Engineering, Indian Institute of Technology Kanpur

June, 2021

- **CPI 7.9**. Coursework included probability, statistics, data structures and algorithms, time series, and advanced statistical methods for analytics.

RECOGNITION & EXTRACURRICULAR

- Awarded with **Next GenAI Leader** in 2025 and **Value Champion Award** in 2024 by *Qure.ai*
- Dramatics Club, IIT Kanpur, 2019–20: Coordinator for 5+ productions with 50+ member team at Antaragni'19 & Inter IIT Cultural Meet
- **JEE Advanced: AIR 943** (2017), **JEE Mains: AIR 1818** (2017), **KVPY Scholar with AIR 643**